

EV Adoption in Washington State:

Structural Patterns and Predictive Analysis of County-Level Adoption

Github(https://github.com/alexsong-lab/SML_group_project/tree/main/final%20project)

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Abstract

This project examined patterns of electric vehicle (EV) adoption across Washington State's 39 counties using both clustering and predictive modeling techniques. Data from the Washington Department of Licensing, the Alternative Fuels Station Locator, and the U.S. Census Bureau were combined to create 16 features related to EV ownership, charging infrastructure, vehicle characteristics, and demographics. PCA reduced the feature set to three principal components explaining 81% of the total variance. K-Means clustering identified three county groups: high-adoption urban counties, a larger rural group, and Garfield County as a unique outlier. The clustering results were supported by hierarchical clustering (ARI = 0.90). Among the supervised learning models, gradient boosting performed best ($R^2 = 0.895$, RMSE = 21,520). Across the strongest-performing models, charging infrastructure and median income emerged as the most important predictors of EV adoption. Overall, the findings suggest that counties face different barriers to EV adoption and may benefit from more targeted policy approaches

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1.0 Introduction

Washington State has one of the highest rates of electric vehicle (EV) adoption in the United States, yet adoption varies substantially across counties. King County has approximately 894,695 registered EVs, while Garfield County has only 34. This striking disparity motivated an investigation into the factors that influence EV adoption, and the objective of this project is to identify and analyze these county-level differences using data science and machine learning methods. Understanding these differences is important because a statewide approach based on average conditions may not adequately address the unique challenges faced by different counties, where urban areas often have higher incomes, greater population density, and transportation needs that differ from those of rural agricultural communities. Clarifying these patterns can help policymakers make more informed decisions about charging infrastructure investments and EV support programs.

Data Sources and Dataset

Data were compiled from three primary sources. EV registration data from the Washington Department of Licensing included vehicle type (battery electric vehicle [BEV] or plug-in hybrid electric vehicle [PHEV]), make, model year, and electric range. Charging infrastructure data from the Alternative Fuels Station Locator provided county-level counts of Level 1, Level 2, and DC fast charging stations. Median household income estimates for 2025 came from the U.S. Census Bureau's American Community Survey. After cleaning and standardization, these records were merged into a unified county-level dataset with derived features characterizing charging availability and adoption patterns. The final dataset consisted of 39 county-level observations and 16 features describing EV ownership, vehicle characteristics, charging infrastructure, and demographic conditions, and served as the foundation for the subsequent unsupervised and supervised learning analyses.

Analytical Methods

Two categories of machine learning methods were used. Unsupervised techniques—Principal Component Analysis (PCA), Singular Value Decomposition (SVD), K-Means clustering, and hierarchical clustering—were applied to identify natural groupings among counties and the primary dimensions driving variation in EV adoption. Supervised methods—Ordinary Least Squares (OLS), Ridge, and Lasso regression, regression trees, bagging, random forests, and gradient boosting—were then used to predict total EV counts from infrastructure and demographic characteristics and to assess the relative importance of different predictors. Comparing multiple approaches allowed us to identify the methods that best captured the relationships in the data while highlighting the factors most strongly associated with county-level EV adoption.

2.0 Theoretical Background

The following theoretical background draws primarily from *An Introduction to Statistical Learning with Applications in Python* [1].

2.1 Principal Component Analysis and SVD

Principal Component Analysis, or PCA, is an unsupervised learning technique used to reduce the dimensionality of a dataset while preserving as much variation as possible. PCA transforms the

original correlated variables into a smaller set of uncorrelated variables called principal components (PC), which are ordered according to the amount of variance they explain in the data. Each principal component is a normalized linear combination of the original variables, expressed as:

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \cdots + \phi_{p1}X_p$$

The coefficients defining this combination are called loadings, and the projected values of each observation onto these directions are called scores. The first principal component captures the greatest amount of variability, with each subsequent component accounting for the largest remaining variance while remaining orthogonal to the previous components. PCA can be understood through two equivalent interpretations: maximizing the variance of the projected data, or minimizing the reconstruction error relative to the original observations.

Because PCA is sensitive to variable scale, the data were standardized to have mean zero and standard deviation one prior to analysis. To assess how much information each component captures, we examined the proportion of variance explained (PVE) and used a scree plot to identify an “elbow” to guide component selection.

Singular Value Decomposition (SVD) is a matrix factorization approach that decomposes a data matrix into orthogonal components representing its underlying structure. In our analysis, the decomposition followed the standard form:

$$X = UDV^T$$

Where V contains the principal component loading vectors, U contains the standardized scores, and D contains the singular values (the standard deviations of the principal components). When applied to standardized data, SVD produces results that are mathematically equivalent to PCA. PCA was used to identify the major dimensions underlying county-level differences in EV adoption, while SVD served as a complementary check on the stability of the PCA solution.

2.2 Clustering Methods

Clustering groups observations based on similarity in multivariate space. K-Means partitions data into k clusters by minimizing within-cluster variation. Because the algorithm can converge to local optima, it is standard to run multiple random initializations and evaluate solutions using elbow plots or silhouette scores. Hierarchical clustering builds a dendrogram showing how observations merge step-by-step. Linkage methods—single, complete, average, and Ward—define how distances between clusters are computed. Ward linkage minimizes increases in within-cluster variance and typically produces compact, interpretable clusters, especially when applied to standardized PCA-transformed data. In this analysis, clustering was performed on the PCA-transformed standardized feature matrix to ensure that distance calculations reflected the major underlying dimensions of variation.

2.3 Regularized Regression

Regularized regression improves prediction by shrinking coefficient estimates and reducing variance. Ridge regression adds an ℓ_2 penalty to the residual sum of squares, minimizing:

$$RSS + \lambda \sum_{j=1}^p \beta_j^2$$

This shrinks coefficients toward zero without eliminating predictors and is effective when many predictors contribute small effects or when multicollinearity is present.

Lasso regression uses an ℓ_1 penalty, minimizing:

$$RSS + \lambda \sum_{j=1}^p |\beta_j|$$

The ℓ_1 penalty can shrink some coefficients exactly to zero, performing variable selection and producing sparse, interpretable models. Both methods require selecting a tuning parameter λ , typically via cross-validation over a grid of candidate values. Stability of selected predictors can be assessed through repeated cross-validation, which evaluates whether the same variables are consistently chosen across resampled training sets. Because ridge and lasso penalties depend on the scale of the predictors, all variables were standardized prior to applying these regularization methods.

2.4 Tree-Based Methods

Tree-based methods are nonparametric machine learning approaches that model relationships in data by recursively partitioning the predictor space into smaller more homogenous regions. Regression trees make predictions by repeatedly splitting the data based on predictor values to minimize variability within each group. Because they can capture nonlinear relationships and interactions among variables, tree-based methods are often useful when traditional linear models fail to adequately represent the underlying structure of the data.

Ensemble methods improve upon single decision trees by combining the predictions of multiple models. Bagging, or bootstrap aggregation, reduce prediction variance by fitting many trees to bootstrap samples of the training data and averaging their predictions. Random Forests extend this approach by additionally selecting a random subset of predictors at each split. This increases diversity among trees and improving generalization performance. Gradient boosting takes a sequential approach, where each new tree is trained to correct the residual errors of the previous trees allowing the model to iteratively improve predictive accuracy.

3.0 Methodology

3.1 Data Preparation

Raw count variables like EV registrations and charger counts were log-transformed to reduce the right skew caused by King County's extreme values. All features were standardized to zero mean and unit variance prior to PCA and regularized regression. Tree-based methods are scale-invariant but were also used the same feature set for consistency. The following derived features were engineered: chargers per 1,000 EVs (infrastructure coverage ratio), BEV-to-PHEV ratio, fast

charger share (proportion of DC fast chargers), and average vehicle age derived from model year. Average electric range had approximately 44% missing values at the county level; county-level observed means were used rather than imputation, which would have introduced noise into an already small dataset.

A critical data leakage issue was identified and corrected during the supervised learning setup. BEV count and PHEV count were initially included as predictors with total EV count as the target. Because these two variables sum directly to the target, every model achieved R^2 of 1.0. This did not indicate strong predictive performance instead; it occurred because the target variable could be derived directly from the predictor variables. Both columns and their log-transformed versions were removed from the supervised feature set. The final feature set for supervised learning comprised 11 variables: log-transformed charger counts (total, Level 1, Level 2, DC fast, and chargers per 1,000 EVs), average model year, average electric range, median income 2025, percent BEV, BEV-to-PHEV ratio, and fast charger share.

3.2 Unsupervised Learning

PCA was applied to the standardized 16-feature matrix used for dimensionality reduction and EDA. The 16 features used for PCA consisted of eight log-transformed count variables (total EVs, BEVs, PHEVs, total chargers, Level 1 chargers, Level 2 chargers, DC fast chargers, chargers per 1,000 EVs) and eight behavioral or demographic variables (percent BEV, percent PHEV, BEV-to-PHEV ratio, average model year, average vehicle age, average electric range, median income 2025, and fast-charger share). The number of components was selected based on a cumulative variance threshold of 80%. SVD was applied to the same matrix, and the resulting directions were aligned with PCA loadings via sign correction. For K-Means clustering, k was varied from 2 to 9 and within-cluster inertia was plotted to identify the elbow. The final model used $k = 3$, $n_init = 20$, and a fixed random seed. Hierarchical clustering was performed on the PCA scores using four linkage methods which were Ward, Complete, Average, Single. All hierarchical clustering was performed using Euclidean distance, consistent with standard practice for PCA score space.

3.3 Supervised Learning

a. Regularized Regression

For the regularized regression models, we used the cleaned county-level dataset and selected only numeric predictors, removing log-transformed features to avoid redundancy. The response variable for both Ridge and Lasso was `total_EV_count`. We tuned both models using cross-validation over a shared logarithmic alpha grid (10^{-4} to 10^4), allowing the penalty strength to be chosen based on out-of-sample performance rather than manual selection. To assess how consistently Lasso selected predictors, we ran a repeated 5 x 20 cross-validation stability check.

b. Tree-based Method

The same county level dataset and supervised feature set described previously were used for all tree-based models. The response variable was total_EV_count. A regression tree was first fit as a baseline model, with max_depth and max_leaf_nodes selected using GridSearchCV. Bagging and random forest models were then implemented to improve predictive performance by aggregating multiple decision trees. For the random forest models, the number of predictors considered at each split (max_features) was selected using cross-validation. Gradient boosting models were evaluated across combinations of learning rates and number of trees using a 5x5 parameter grid. Model performance was assessed on the held-out test set using R^2 and RMSE. Variable importance measures from the random forest and gradient boosting models were used to identify predictors most strongly associated with county level EV adoption.

4.0 Computational Results

4.1 PCA

Table 1. Principal component summary

COMPONENT	VARIANCE EXPLAINED	CUMULATIVE	DOMINANT POSITIVE LOADINGS	NEGATIVE LOADINGS
PC1	49.3%	49.3%	EV registrations, chargers, income	%PHEV, vehicle age
PC2	19.0%	68.3%	electric range, model year, %BEV	%PHEV, vehicle age
PC3	12.7%	81.0%	fast charger share, chargers per 1,000 EVs	—

Explanation. The first three components account for 81.0% of total variance, and the PCA solution matches the SVD decomposition exactly (PC1–V1 = 1.0000; PC2–V2 = 1.0000).

4.2 Clustering

Table 2. Summary of clustering results

ITEM	VALUE
OPTIMAL K (ELBOW)	3 (inertia 330.2 → 240.2)
CLUSTER 0 SIZE	11 counties
CLUSTER 0 MEAN EVS	145,343
CLUSTER 0 MEAN INCOME	\$102,363
CLUSTER 0 CHARGERS / 1,000 EVS	3.17
STABILITY — WARD ARI	0.900
STABILITY — COMPLETE ARI	0.557
STABILITY — AVERAGE / SINGLE ARI	0.182
GARFIELD COUNTY	Isolated cluster across all methods

Explanation. A three-cluster solution was selected at the elbow point, with Cluster 0 containing 11 counties and Ward linkage showing the strongest agreement (ARI = 0.900), while Garfield County consistently formed a singleton cluster.

4.3 Supervised Learning

a. Regularized Regression Results

Table 3. Summary of Ridge and Lasso Regression Results.

METRIC	RIDGE	LASSO
BEST A	0.000100	0.000175
TRAIN RMSE	0.0000	0.0013
TEST RMSE	0.0001	0.0010
TRAIN R ²	1.0000	1.0000
TEST R ²	1.0000	0.9999
NON-ZERO COEF.	15	4
TOP COEF.	BEV_count, PHEV_count, percent_BEV, percent_PHEV, BEV_to_PHEV_ratio	BEV_count, PHEV_count, median_income_2025, BEV_to_PHEV_ratio
REPEATED CV RMSE	—	0.0077 ± 0.0247

Explanation. Table 3 reports the selected regularization parameters, training and test RMSE and R², the number of non-zero coefficients, the highest-magnitude coefficients, and the repeated cross-validation RMSE for each model.

Stability. Across repeated cross-validation, Lasso selected BEV_count and PHEV_count in 100/100 folds and median_income_2025 in 93/100 folds; all other predictors were selected in fewer than 40/100 folds.

b. Tree-Based Methods Results

Table 4. Summary of Tree-Based Methods Results.

MODEL	TEST R ²
DECISION TREE	-9.95
BAGGING	0.810
RANDOM FORREST	0.782
GRADIENT BOOSTING	0.895

Explanation. Table 4 reports the held-out test R² for each tree-based model. Gradient boosting produced a test RMSE of 21,520 using a final configuration of 5,000 trees and a learning rate of 0.001. In the random forest model, the most influential predictors were Level 2 chargers, total chargers, median household income, DC fast chargers, the BEV-to-PHEV ratio, and the percentage of BEVs.

5.0 Discussion

5.1 Unsupervised Results

The PCA results suggest that there are meaningful differences between counties in Washington rather than patterns created by the analysis itself. Since PC1 alone explained 49.3% of the total variance, much of the variation across counties can be described by a single underlying factor related to the overall size and development of the local EV ecosystem. Beyond this dominant dimension, PC2 can be viewed as representing vehicle fleet modernity, driven by newer model years, longer electric range, and a higher share of BEVs, while PC3 reflects differences in charging infrastructure availability that are not simply related to county size.

The counties grouped in Cluster 0—King, Pierce, Snohomish, Clark, Spokane, Kitsap, Thurston, Whatcom, Benton, Skagit, and Chelan—are some of the state's most populated and economically developed areas, largely concentrated in western Washington and around Spokane. These counties tend to have higher incomes, larger EV populations, and more charging infrastructure. One interesting finding was that Cluster 0 had the lowest charger-to-EV ratio (3.17 compared with 7.29 in Cluster 1). Although these counties have the largest charging networks overall, EV adoption has grown so quickly that infrastructure expansion has struggled to keep pace with demand. The three-cluster structure was also robust: Ward linkage reproduced the K-Means solution almost exactly (ARI = 0.900), differing only in the placement of Kittitas County.

Garfield County stood out in every clustering method we tested. It had only 34 registered EVs, no public charging stations, the highest share of PHEVs, the shortest average electric range, and the lowest median income among all counties in the dataset. Rather than appearing as a lower-adoption version of the rural counties, Garfield consistently emerged as its own group. This suggests that the factors affecting EV adoption there may be different from those in other parts of the state. Limited charging availability, lower income levels, and the transportation needs associated with agricultural and rural communities may all contribute to its unique profile.

5.2 Supervised Results

Building on the structure identified through PCA and clustering, the supervised models were then used to predict county-level EV counts and to rank predictor importance. The linear models generally performed poorly, which is likely due to the small training sample of only 29 counties and the outsized influence of King County. With nearly 895,000 registered EVs, King County is far larger than any other county in the dataset, making it difficult for traditional linear models to capture the underlying relationships. Depending on whether King County appeared in the training or test set, the models either fit too closely to that extreme value or struggled to make accurate predictions. This is reflected in the OLS test R^2 of -2.919, which indicates that simply predicting the average EV count would have produced better results. The large Ridge penalty ($\alpha = 10,000$) further suggests that the model had difficulty finding a stable linear relationship in the data.

In contrast, gradient boosting achieved a test R^2 of 0.895, making it the strongest-performing model in this study. Tree-based ensemble methods such as gradient boosting and random forests are better suited for datasets with non-linear relationships and extreme observations because they can split the data into smaller, more homogeneous groups. As a result, they were able to capture patterns

that the linear models missed. Both random forest and gradient boosting identified charger infrastructure and median income as the most important predictors, which increases confidence that these variables play a meaningful role in explaining EV adoption across counties.

One result that remained consistent across the linear models was the importance of income. In the Lasso stability analysis, median income was selected in all 100 cross-validation folds, making it the most reliable predictor identified by the linear methods. This finding is also consistent with the clustering results, where higher-adoption counties generally had higher incomes. However, income should be interpreted carefully because it is likely correlated with other factors such as population density and urbanization, making it difficult to separate its independent effect on EV adoption.

5.3 Limitations

This study has several limitations that should be considered when interpreting the results. First, the dataset includes only 39 counties, which is a relatively small sample for supervised learning. Because the test set contained just 10 counties, model performance estimates can vary substantially depending on which counties were included in the split. For this reason, the cross-validation results likely provide a more reliable assessment of model performance than a single train-test split, and the strong results achieved by gradient boosting should be viewed cautiously.

Second, while median income consistently emerged as an important predictor, it is likely related to other factors such as population density and urbanization. As a result, it is difficult to determine whether income itself is driving EV adoption or simply reflecting broader characteristics of more urban counties.

Another limitation is that the analysis uses data from a single point in time. Since the dataset is cross-sectional, it cannot capture changes in EV adoption over time or determine whether lower-adoption counties are gradually catching up or facing longer-term barriers to adoption.

Finally, the analysis was conducted at the county level, which masks important differences within individual counties. For example, residents living in apartments or multi-family housing may face charging challenges that are not reflected in county-level averages for income or charging infrastructure. As a result, some local factors influencing EV adoption are not captured in the current dataset.

6. Conclusions

This study set out to explain why EV uptake differs so sharply from one county to the next, treating that variation as the central question rather than the overall statewide total. By pairing unsupervised pattern discovery with predictive modeling, we surfaced several consistent patterns that help account for these differences.

The clustering analysis showed that the 11 counties in Cluster 0, most of which are located along the I-5 corridor, have already developed strong EV ecosystems and are likely to continue seeing growth in EV adoption. However, these counties also have the lowest charger-to-EV ratios, suggesting that charging infrastructure has not expanded as quickly as EV ownership. Additional

investment in charging networks and supporting infrastructure may help meet future demand in these areas.

The 27 counties in Cluster 1 face a different situation. While they generally have enough charging infrastructure relative to their current EV populations, lower income levels may make EV adoption more difficult. Programs that improve affordability, such as purchase incentives or access to lower-cost used EVs, may be more effective than simply adding more charging stations.

Garfield County consistently appeared as its own cluster and differed substantially from every other county in the dataset. Its extremely low EV adoption, lack of public charging stations, and lower income levels suggest that the challenges it faces may not be addressed by the same strategies that work in other counties.

The supervised learning results supported many of the same conclusions. Across the models that performed well, charging infrastructure and median income were the strongest predictors of EV adoption. Together, these findings suggest that different counties may benefit from different policy approaches rather than a single statewide strategy.

There are several opportunities for future research. Using data collected over multiple years would help determine whether some counties are gradually increasing EV adoption or facing longer-term barriers. Additional variables such as population-adjusted measures, housing characteristics, and vehicle use patterns could provide a better understanding of EV adoption in rural communities. Expanding the analysis to other states would also help determine whether the patterns observed in Washington are present elsewhere.

7. References

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